L2.4 Limits at Infinity

§3.4 — where the asymptote rules come from

The three asymptote rules

① BOT > TOP

HA $y = 0$

e.g. $\dfrac{x^2 + x}{x^3}$

② BOT = TOP

HA at the ratio of leading coefficients

e.g. $\dfrac{3x^2 - 1}{4x^2}$

③ TOP > BOT

no HA; SA by polynomial long division

e.g. $\dfrac{4x^2 - 4}{3x - 6}$

THE LIMIT AT INFINITY THEOREM — all three rules run on this one

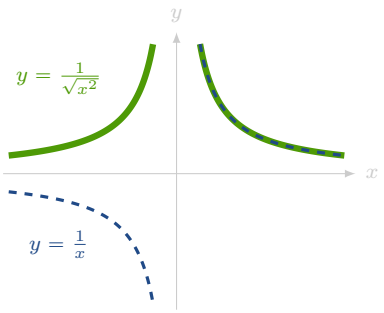
$$\lim_{x\to\infty} \frac{a}{x^r} = 0 \qquad a \text{ real}, \quad r \text{ a positive rational}$$

Also true as $x \to -\infty$, whenever x^r is defined there.

Ex. Evaluate $\displaystyle\lim_{x\to\infty} \frac{3x^2 - 1}{4x^2 + x}$.

$$\lim_{x\to\infty} \frac{3x^2 - 1}{4x^2 + x} = \lim_{x\to\infty} \frac{(3x^2 - 1)/x^2}{(4x^2 + x)/x^2} = \lim_{x\to\infty} \frac{3 - \frac{1}{x^2}}{4 + \frac{1}{x}} = \frac{3 - 0}{4 + 0} = \boxed{\frac{3}{4}}$$

$\sqrt{x^2} = |x|$ — the sign on the way to $-\infty$



A square root is never negative, so $\sqrt{x^2}$ is not x — it is $|x|$.

$$x < 0 \implies \frac{1}{x} = -\frac{1}{\sqrt{x^2}}$$

"on the way to $-\infty$ the $\frac{1}{x}$ is negative, and the radical is not, so a minus sign has to appear when you move it inside"

Ex. Evaluate $\displaystyle\lim_{x\to-\infty} \frac{\sqrt{4x^2 + 1}}{x - 3}$.

Divide top and bottom by x. Since $x < 0$, use $\frac{1}{x} = -\frac{1}{\sqrt{x^2}}$ on top.

$$\lim_{x\to-\infty} \frac{\sqrt{4x^2 + 1}}{x - 3} = \lim_{x\to-\infty} \frac{-\sqrt{4x^2 + 1}/\sqrt{x^2}}{(x - 3)/x} = \lim_{x\to-\infty} \frac{-\sqrt{4 + \frac{1}{x^2}}}{1 - \frac{3}{x}} = \frac{-\sqrt{4 + 0}}{1 - 0} = \boxed{-2}$$

SLANT ASYMPTOTE — and why the line is there

Ex. Find the slant asymptote of $f(x) = \dfrac{4x^2 - 4}{3x - 6}$.

Polynomial long division:

$$\begin{array}{r|l} & \frac{4}{3}x + \frac{8}{3} \\ \hline 3x - 6 & 4x^2 - 4 \\ & \underline{-4x^2 + 8x} \\ & 8x - 4 \\ & \underline{-8x + 16} \\ & 12 \end{array}$$

$$\frac{4x^2 - 4}{3x - 6} = \underbrace{\tfrac{4}{3}x + \tfrac{8}{3}}_{\to +\infty} + \underbrace{\frac{12}{3x - 6}}_{\to 0} \qquad SA = \tfrac{4}{3}x + \tfrac{8}{3}$$

"the remainder going to zero is the whole reason the line is an asymptote — the limit is $+\infty$ and the answer is still a line"

$\infty - \infty$ — rewrite first

∞ is not a number, so the limit laws do not apply to it. Rewrite, *then* take the limit. For a difference of two roots, or a root minus a polynomial, multiply by the conjugate over itself — you have done conjugates before.

Ex. Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+1} - x\right)$.

$$\begin{aligned} \lim_{x\to\infty} \left(\sqrt{x^2+1} - x\right) &= \lim_{x\to\infty} \left(\sqrt{x^2+1} - x\right) \cdot \frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1} + x} \\ &= \lim_{x\to\infty} \frac{(x^2+1) - x^2}{\sqrt{x^2+1} + x} \\ &= \lim_{x\to\infty} \frac{1}{\sqrt{x^2+1} + x} \\ &= \boxed{0} \end{aligned}$$